GAN HR Aptitude Test

Study Guide

GAN
HUMAN RESOURCES

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How to Use This Guide

Use this study guide to prepare for your testing session. An overview of each test is provided, including general rules, guidelines, definitions, and practice problems/answers to better prepare you for the test. We hope you find this information helpful as you prepare for your testing session.

General Testing Information

- The tests have been carefully developed to ensure that they are fair to all applicants. All applicants have an equal chance of doing well, regardless of their background or prior experiences.
- The tests were developed by psychologists who are experts at evaluating the results to determine apprenticeship potential.
- This study guide was developed to align with the tests. If you choose to review the study guide and practice the essential skills taught, you will enhance your overall performance on the test.
- You do not need experience or special knowledge to succeed on the tests.
- The tests measure your ability to think, solve problems, and be successful in the apprenticeship.

Test-Taking Tips

Before the Test

- Stay positive! A positive mindset will reduce your stress on a test and lead to better chances of success.
- Take notes on what you are learning - and then go back and review them.
- Practice, practice, practice. The more questions you answer, the more comfortable you will be on the test.
- Practice relaxing and staying calm. Strategies, such as deep breathing or imagining yourself in a relaxing place, will help you calm down if you get nervous during the test.
- The night before the test:
    - Review your notes and any skills you are particularly nervous about. This is the time to refresh your memory, not learn something new.
    - Make sure to set your alarm (or multiple alarms if you are a heavy sleeper!).
    - Go to bed early. Keep your phone usage and TV watching to a minimum. This will help you sleep better and also ensure the notes you have reviewed stay fresh in your mind.

Test-Taking Tips

On Testing Day

- Make sure to eat a healthy, not heavy, breakfast.
- Plan to arrive at the testing site at least 30 minutes early. Doors close at the start of the test and there is no late admittance. Assume there will be traffic, so that if there is traffic, you are not late.
- Review your notes as you wait for the test to start. Again, this isn't about learning new skills, but about making sure what you've already learned is fresh in your mind.
- Stay positive! If you're reading this study guide, you're preparing for the test. If you're preparing for the test, you're giving yourself the tools to succeed. Remember: **You have the tools to succeed.**

Taking the Test

- Listen to all the directions from the test administrator.
- Read all the directions and examples on the test.
- Regularly check the number on your answer sheet and make sure it matches the question number from the test booklet.
- Make sure to budget your time. Every question is worth the same amount; do not spend too long on any one question.
- If you feel a question is too hard, skip it and answer the easier ones first. Do not forget to return back to the questions you skipped!
- For each question:
    - Make sure to read the entire question carefully.
    - Try to predict the correct answer ***before*** looking at the answer choices.
    - Read ***each*** possible answer, even if you think you have already found the correct answer.
    - If you're unsure of the correct answer, eliminate any obviously wrong answers.
    - Choose the best answer.
- If you have extra time, double check your answers. Make sure the answer you marked on the answer sheet is the answer you meant to mark.
- If you change an answer, make sure you erase your old answer completely.
- Erase any stray marks on the answer sheet.

Mechanical Comprehension

Overview

The mechanical comprehension test measures how well you understand the relationships between physical forces and mechanical elements in authentic situations. The problems in this test represent simple, frequently encountered mechanisms that do not require special mechanical knowledge or training to solve.

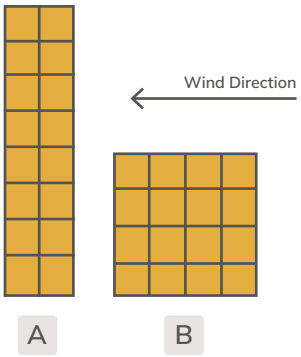
Test-Taking Tips

- Rely on your prior knowledge and common sense to solve these problems. Many of them will be similar to situations you have experienced in your day-to-day life.
- Remember that there are three options for each answer.
- Try to visualize the problem in 3D and think about what would happen in real life. For example, if the problem depicts two cars driving, visualize the two cars actually driving.

Examples

1 Center of Gravity

Which tower of boxes is more likely to fall over from the wind?



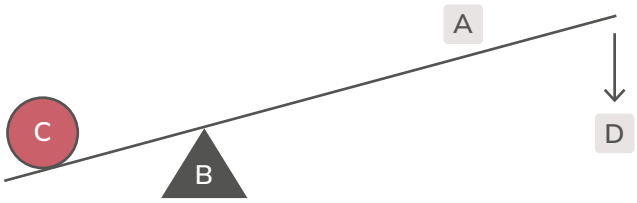
Step 1	Look at the shape and size of each tower of boxes.	Notice that tower A is 2 boxes wide and 8 boxes tall and tower B is 4 boxes wide and 4 boxes tall.
Step 2	Think about your past experiences with stacking items. What types of stacks had more stability?	The lower the center of gravity, the more stable an object, or stack of objects, will be. Tower A, since it is taller, has a higher center of gravity (box level 4). Tower B, since it is shorter, has a lower center of gravity (box level 2). Therefore, tower A will be more likely to fall over from a strong wind gust.

Real World Connections

- stacking objects
- an ice skater completing a triple axle (spinning while jumping in the air) keeps his/her arms to his/her side to lower his/her center of gravity
- when taking a curve quickly, a sedan is less likely to tip over than an SUV, as the sedan has a lower center of gravity
- choosing a doorstop from between two similarly weighted options - you would choose the doorstop that is lower to the ground, rather than one that is taller

2 Torque

At which point (A, B, or D) would it be easiest to move object C?



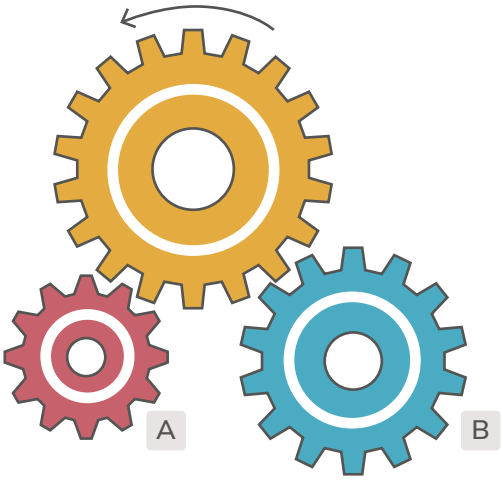
Step 1	Identify where each point is located relative to object C in the figure.	Note that: • Point B is closest to C, • Point D is farthest from C, • Point A is between B and D.
Step 2	Consider which point would offer the most leverage to move object C in a real scenario.	Point D provides the most leverage, as it is the furthest away from the point we need to move. Therefore, it will be easiest to move object C at point D.

Real World Connections

- a child playing on a seesaw
- using a longer handle vs a shorter handle shovel to dig a hole
- carrying a large/long object (like a ladder or a beam) where both people would hold the object at the opposite ends, as opposed to closer to the middle.

3 Gears

Which gear will rotate faster?



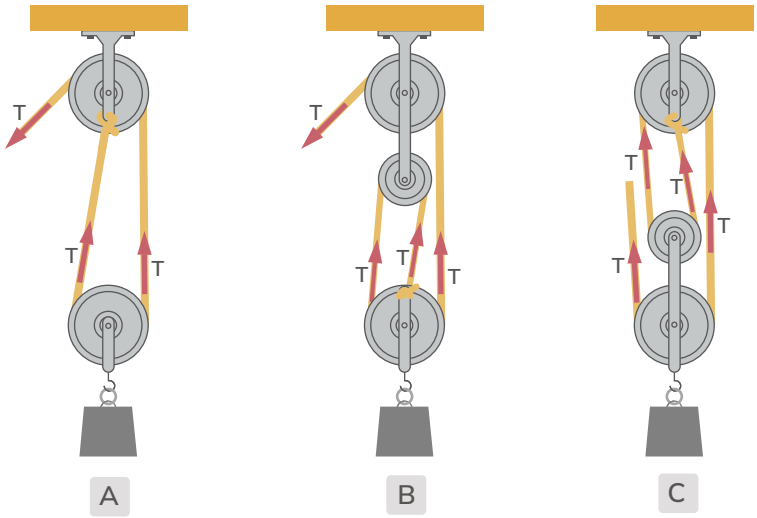
Step 1	Since this question is referring to the speed, we need to take note of the size of the gears, not the direction of rotation.	Gear A is smaller than gear B and has less teeth.
Step 2	Think about how objects move in real life situations.	Smaller objects have less distance to travel for a full rotation. Therefore, the smaller the gear, the faster it will complete a full rotation. Since gear A is smaller than gear B, gear A will rotate faster.

Real World Connections

- the back wheels on a tricycle, which are smaller than the front wheel, rotate much faster than the larger front wheel
- all other things being equal, a smaller diameter screw will penetrate the object faster than the larger diameter screw
- center lug nut of a Formula One race car will spin faster than the rubber tire

4 Pulleys

Which pulley will require the least amount of work to lift the weight?



Step 1	Look at the number of cables and pulleys in each diagram.	Diagram A has two pulleys and two cables pulling force upwards. Diagram B has three pulleys and three cables pulling force upwards. Diagram C has three pulleys (just like diagram B) but it has four cables pulling force upwards.
Step 2	Think about how the weight of the object being lifted is being distributed.	In diagram A, the weight is being split between the two cables. In diagram B, the weight is being split between three cables. In diagram C, the weight is being split between four cables. Since diagram C has the most cables splitting the weight, it will require the least amount of work to lift it.

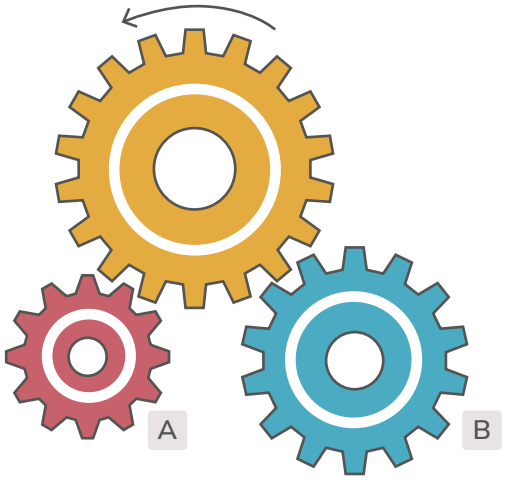
Real World Connections

- Blinds for windows are operated by pulleys. When the window is wider and the blinds are heavier, there are more strings to raise or lower the blinds.
- Roofers bringing shingles onto the roof would use a pulley system with multiple pulleys and cables to lighten the load.
- A pulley system at the top of the elevator shaft helps lift and lower the elevator car.

Practice Test

This is a practice mechanical comprehension test.
You have 8 minutes to complete this test. (18 problems).

1. Which direction will gears A and B turn?

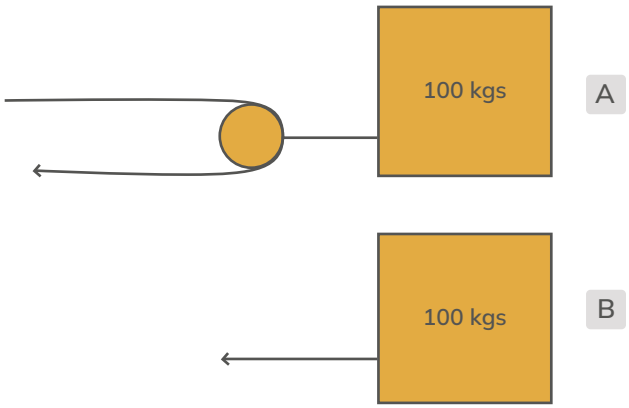


A) Both clockwise

B) Both counter-clockwise

C) Gear A will turn clockwise and gear B counter-clockwise

2. Which object requires less force to move in the direction of arrow?

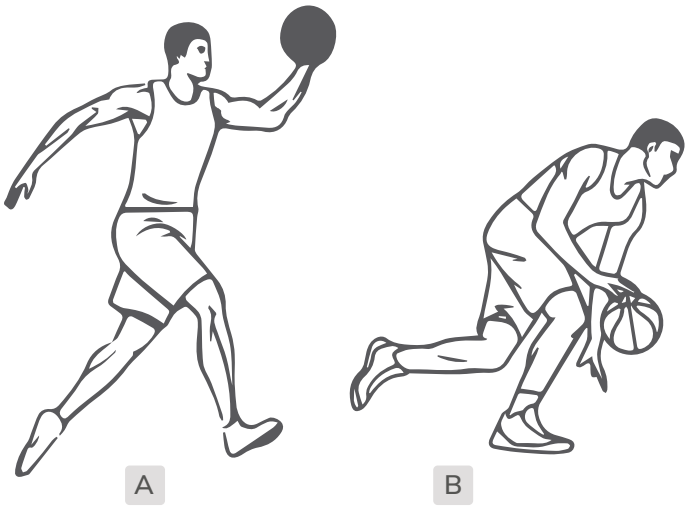


A) Object A

B) Object B

C) They both require equal force

3. Which basketball player is more likely to retain control of the ball?

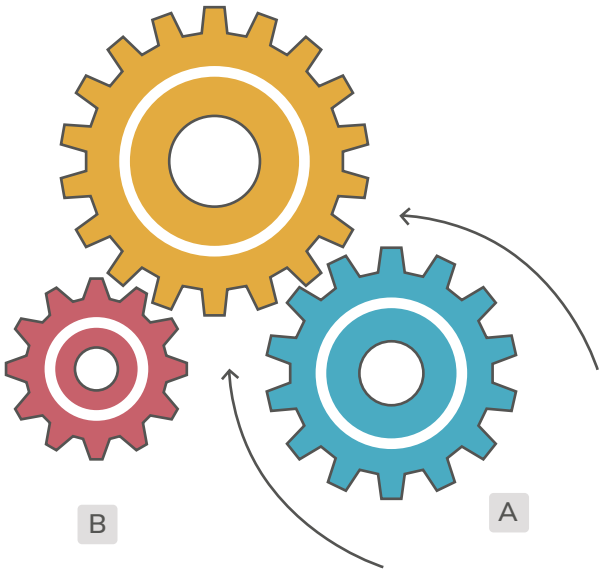


A) Player A

B) Player B

C) Both equally likely

4. Which direction does gear A need to rotate so that gear B rotates in a counter-clockwise direction.

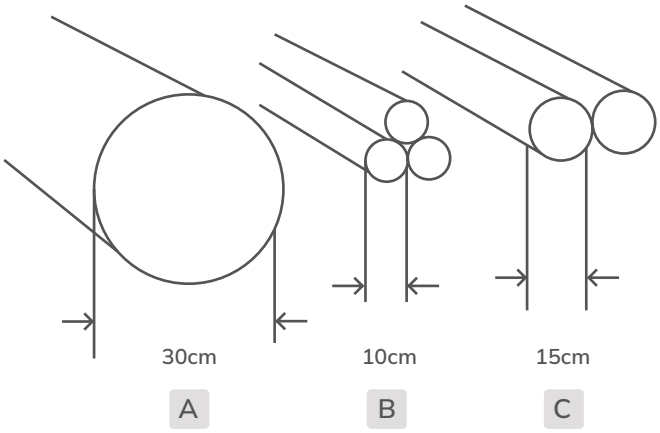


A) clockwise

B) counter-clockwise

C) Gear A will not impact gear B

5. If there is a 100 pound weight on each of the pulleys below, which would require the least force to move?

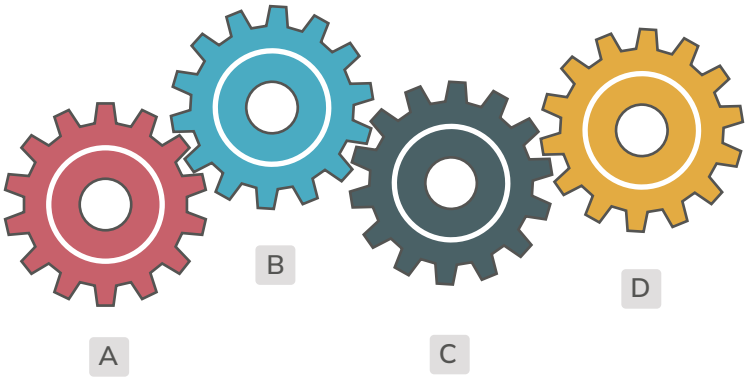


A) Figure A

B) Figure B

C) Figure C

6. If wheel A turns counter-clockwise, in which direction will wheel D turn?

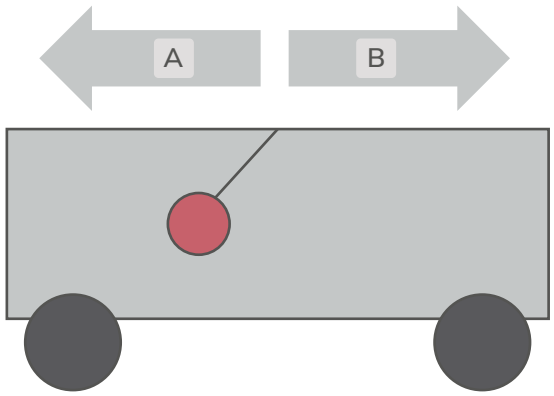


A) clockwise

B) counter-clockwise

C) It will not turn

7. In which direction is the train traveling?

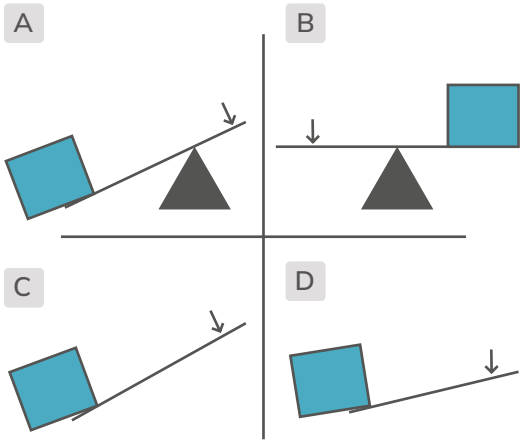


A) Direction A

B) Direction B

C) Neither direction

8. Which figure shows the object in a state of balance?



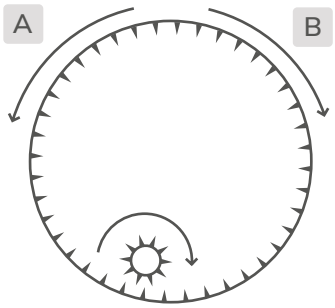
A) Figure A

B) Figure B

C) Figure C

D) Figure D

9. In which direction will the large wheel turn if the small wheel turns in the direction indicated?

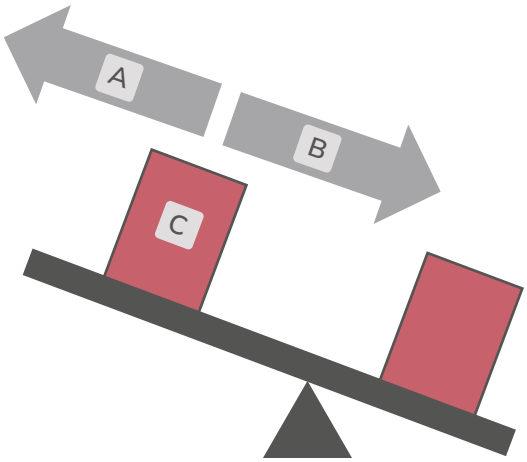


A) Counter-clockwise

B) Clockwise

C) It will not turn

10. In which direction should box C move in order to balance the board?

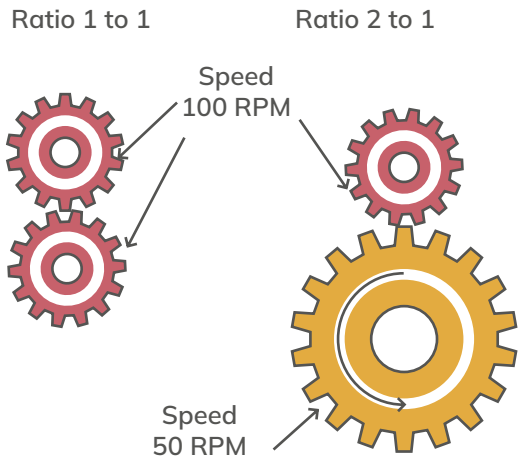


A) Direction A

B) Direction B

C) Neither direction

11. How many times will the large wheel go around if the small wheel makes two revolutions?

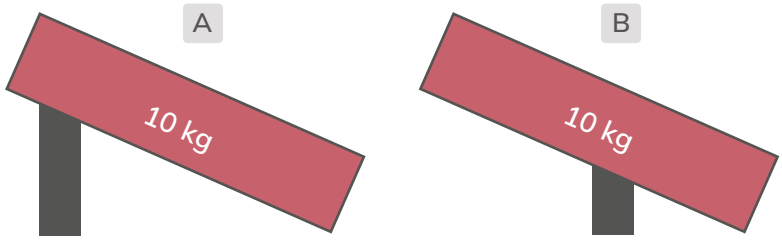


A) 1 revolution

B) 2 revolutions

C) It cannot be determined

12. Which pole bears more weight?

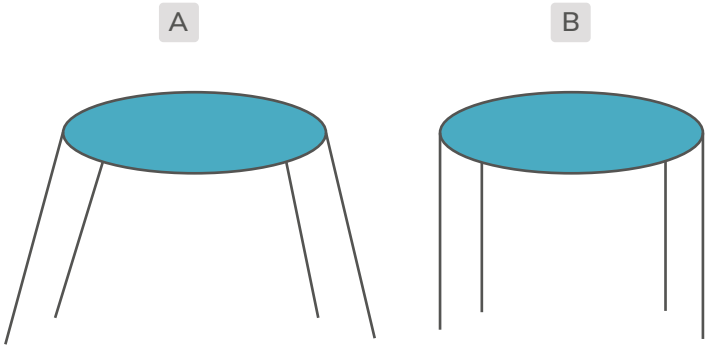


A) Pole A

B) Pole B

C) No difference

13. Which stool would be less wobbly if you rocked while sitting on it?

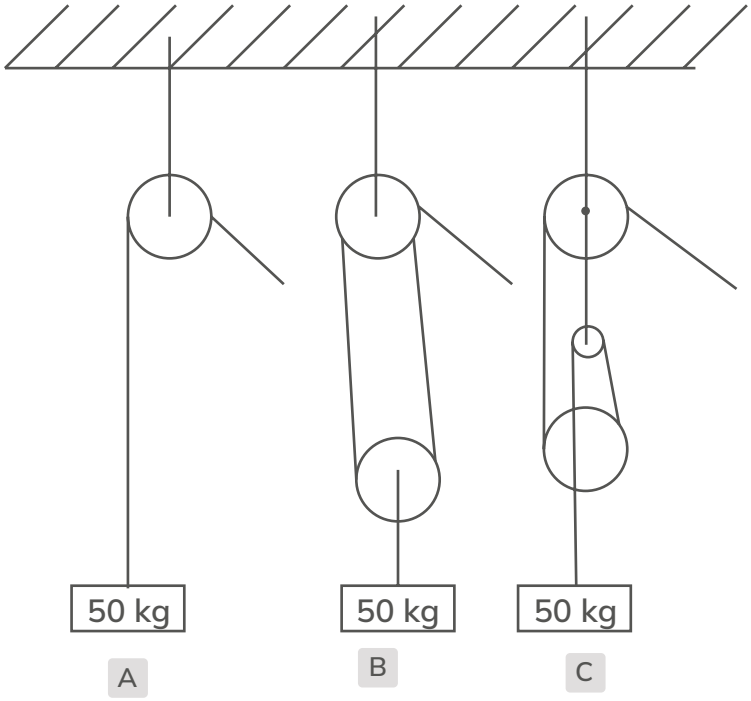


A) Stool A

B) Stool B

C) No difference

14. Which of the pulleys will require the least amount of work to lift the weight?

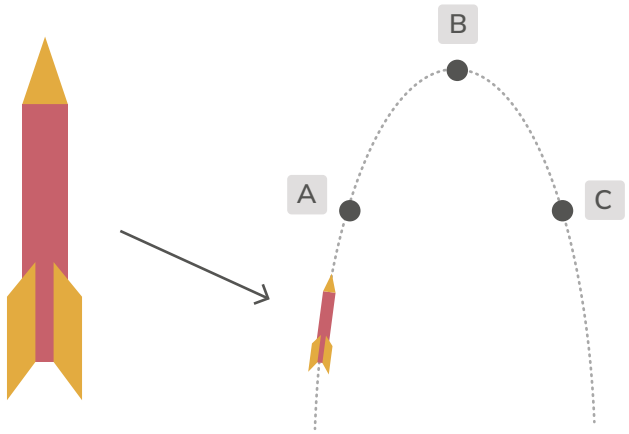


A) Pulley A

B) Pulley B

C) Pulley C

15. At which point will the rocket be at its lowest velocity?

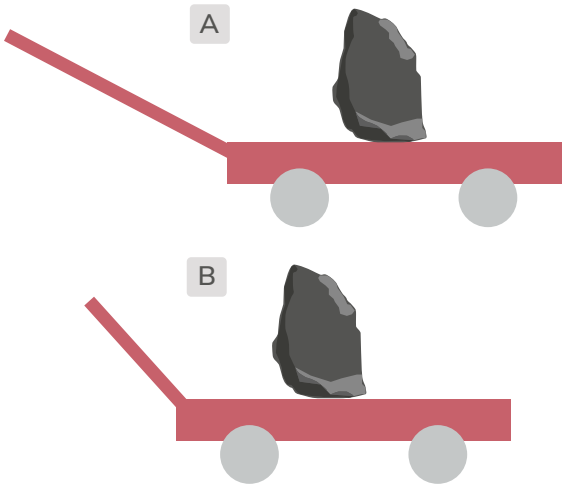


A) Point A

B) Point B

C) Point C

16. Which cart would take less effort to pull?

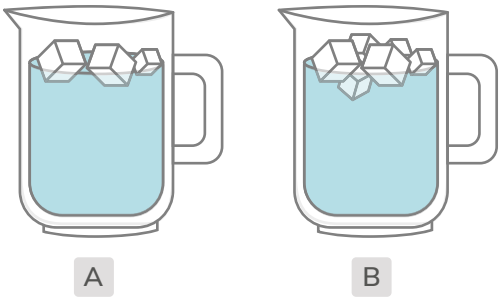


A) Cart A

B) Cart B

C) Both would take the same amount of effort

17. In which pitcher will the ice cubes melt first?

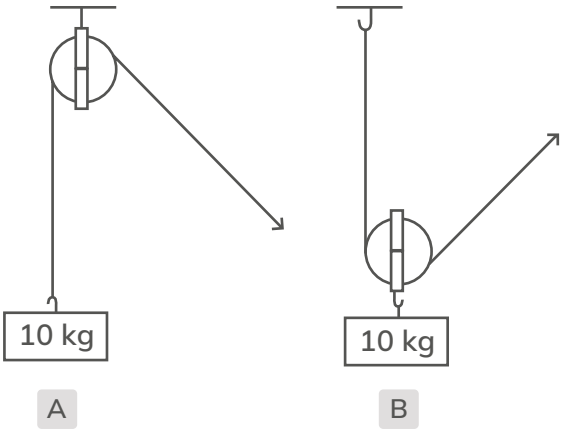


A) Pitcher A

B) Pitcher B

C) They will melt at the same time

18. Which pulley will require less force to lift the object?



A) Pulley A

B) Pulley B

C) They will require the same amount of force

Practice Test Answers

1) A	**2)** A	**3)** B	**4)** B	**5)** A	**6)** A
7) B	**8)** B	**9)** B	**10)** A	**11)** A	**12)** B
13) A	**14)** C	**15)** B	**16)** C	**17)** C	**18)** A

Numerical Computation

Overview

The Numerical Computation test assesses your knowledge of arithmetic involving fractions and decimals. This section of the study guide will review the skills you will need in order to solve these problems.

- Pay attention to the arithmetic operations in the problem.
- Use your scratch paper, but keep your work neat. You do not want to get the answer wrong because your handwriting is illegible or overlaps with another problem.
- Try to estimate the correct answer. If the answer you calculate is far away from the answer you estimate, double-check your calculation.
- If you're working with fractions, make sure to reduce the fraction completely. You might calculate 2/8 as your answer, but you won't find that on the answer sheet. You'll need to reduce it to 1/4 in order to find the answer on the test.
- If the answer you calculated is one of the options, fill in the correct circle on your answer sheet. If it isn't an option, it's generally quicker to redo the problem, rather than try to find your error.

Definitions

- **Fraction:** A fraction is a part of a whole.
- **Numerator:** The numerator is the number on the top of the fraction. The numerator is the number of parts out of the whole.
- **Denominator:** The denominator is the number on the bottom of the fraction. The denominator is the total number of parts.
    - If I cut a pizza into 8 pieces and eat 1 of the pieces the fraction of the pizza that I ate is 1/8. The numerator of this fraction (1) shows the part of the whole that I ate. The denominator (8) shows the total number of pieces.
    - The fraction of the pizza left on the table is 7/8. The numerator of this fraction (7) shows the part of the pizza left on the table. The denominator (8) shows the total number of pieces.
    - Notice that the denominator for both fractions does not change, as the total number of pieces for my pizza was always 8.
- **Proper/Common Fractions:** A proper or common fraction is a fraction that has been fully reduced. The numerator (top number) is smaller than the denominator (bottom number) and there is no number that can be divided evenly out of both the numerator and denominator.
    - 1/2, 3/4, 5/8, etc...

Numerical Computation

Definitions

- **Unreduced Fraction:** An unreduced fraction has a smaller numerator than denominator, just like a proper/common fraction. Unlike a proper/common fraction, with an unreduced fraction, there is a number that can be divided evenly out of both the numerator and denominator. Once you divide this out, the unreduced fraction will become a fully reduced proper/common fraction.
    - 2/4, 6/8, 10/12, etc...
- **Mixed Number:** A mixed number is a whole number with a proper/common fraction.
    - 5 1/2, 7 6/7, 12 2/3, etc...
- **Improper Fraction:** An improper fraction can be reduced to a mixed number. An improper fraction has a larger numerator (top number) than denominator (bottom number). Once you convert the improper fraction to a mixed number, you may still need to reduce the fractional part of your answer.
    - 10/6, 3/2, 9/4, etc...
- **Reducing Fractions:** Reducing fractions means dividing both the numerator and denominator by the same number that goes evenly into both. When you reduce a fraction, you are changing the numerator and denominator, but not the value of the fraction.
    - 2/4 = 1/2, 6/8 = 3/4, etc...
- **Common Denominator:** A common denominator is when you have the same number in the denominator of all the fractions in the problem. We need a common denominator when adding and subtracting fractions. Often, we try to find the lowest common denominator, which is the smallest possible number that all the denominators go into evenly. However, we don't need the lowest common denominator when adding and subtracting fractions, just a common denominator. If you're stuck on finding the lowest common denominator, simply multiply the denominators together to obtain a common denominator.
- **Percents:** A percent, like a fraction, represents part of a whole. When we are dealing with percents, we need to convert the percentage into a decimal. To do this, you move the decimal place two spaces over to the left.
    - 30% = .30, 67% = .67, 8% = .08

Examples & Practice Problems

1 Reducing Fractions

Example

Reduce the fraction 6/12 so that it becomes a common fraction.

Step 1	Find a number that divides evenly into the numerator (6) and the denominator (12). *Tip: When both numbers are even, 2 will always divide evenly.*	6/12 ÷ 2/2 = 3/6
Step 2	Check to see if anything (besides 1) goes evenly into your new numerator and denominator.	3/6 ÷ 3/3 = 1/2
Step	Check to see if anything (besides 1) goes evenly into your new numerator and denominator. *Tip: When your numerator is 1, your fraction is always as reduced as possible.*	Nothing goes evenly into both 1 and 2. Therefore, 6/12 reduces to 1/2.

Practice Problems

Reduce each fraction to simplest terms.

1) 2/4 =	**2)** 35/40 =	**3)** 10/16 =	**4)** 8/36 =	**5)** 18/20 =
6) 4/36 =	**7)** 6/9 =	**8)** 2/10 =	**9)** 3/30 =	**10)** 44/48 =
11) 5/15 =	**12)** 10/35 =	**13)** 10/45 =	**14)** 6/14 =	**15)** 28/32 =
16) 20/24 =	**17)** 5/15 =	**18)** 4/32 =	**19)** 30/35 =	**20)** 3/6 =

Answers on Next Page

Answers to Practice Problems

1) 1/2	**2)** 7/8	**3)** 5/8	**4)** 2/9	**5)** 9/10
6) 1/9	**7)** 2/3	**8)** 1/5	**9)** 1/10	**10)** 11/12
11) 1/3	**12)** 2/7	**13)** 2/9	**14)** 3/7	**15)** 7/8
16) 5/6	**17)** 1/3	**18)** 1/8	**19)** 6/7	**20)** 1/2

2 Common Denominator

Example 1

Find a common denominator between 1/2 and 2/7. Change each fraction so that it has the new denominator.

Strategy: Multiply the denominators to get a common denominator.

Step 1	Multiply the denominators.	2 x 7 = 14. A common denominator between 1/2 and 2/7 is 14.
Step 2	Multiply each fraction by the number that will make that fraction's denominator 14.	1/2 x 7/7 = 7/14 2/7 x 2/2 = 4/14

Example 2

Find the lowest common denominator of 2/3 and 5/36. Change each fraction so that it has the new denominator.

Strategy: Divide the larger denominator by the smaller denominator

Step 1	Divide 36 by 3 to identify what you need to multiply 3 by to get 36 as the denominator of the first fraction.	36 ÷ 3 = 12
Step 2	Multiply 2/3 by 12/12.	2/3 x 12/12 = 24/36

Example 3

Find the lowest common denominator of 1/6 and 7/9. Change each fraction so that it has the new denominator.

Strategy: Find multiples of the larger denominator and then check if the smaller denominator goes into that number.

Step 1	Find the multiples of the larger denominator (9).	9, 18, 27, 36, 45, 54...
Step 2	Check if the smaller denominator (6) divides evenly into one of the multiples you found in step 1.	Does 6 go into 9? No. Does 6 go into 18? Yes. 6 goes into 18 three times. The lowest common denominator is 18.
Step 3	Multiply 1/6 by 3/3.	1/6 x 3/3 = 3/18

Practice Problems

Find a common denominator for all the fractions within the problem and change each so that they all have the same denominator.

1) 2/3, 1/4	**2)** 4/5, 6/7	**3)** 1/2, 3/4	**4)** 2/3, 7/9	**5)** 1/2, 4/5
6) 1/4, 7/8	**7)** 1/10, 1/8	**8)** 4/9, 1/6	**9)** 1/5, 1/9	**10)** 5/7, 2/11
11) 11/12, 1/9	**12)** 5/6, 7/15	**13)** 2/5, 4/13	**14)** 1/2, 8/9	**15)** 3/10, 9/13
16) 1/8, 3/5	**17)** 1/2, 2/3, 3/4	**18)** 1/3, 2/5, 5/12	**19)** 1/5, 2/9, 4/45	**20)** 6/7,1/6, 3/14

Answers to Practice Problems

*Note: These answers are the lowest common denominator. You may have found a different common denominator. Your answer may be different if you found a common denominator, but did not find the **lowest** common denominator. Remember, when adding and subtracting fractions, it is okay if your denominator isn't the lowest common denominator (LCD).*

1) LCD = 12 8/12, 3/12	**2)** LCD = 35 28/35, 30/35	**3)** LCD = 4 2/4, 3/4	**4)** LCD = 9 2/3, 7/9	**5)** LCD = 10 5/10, 8/10
6) LCD = 8 2/8, 7/8	**7)** LCD = 40 4/10, 5/40	**8)** LCD = 18 8/18, 3/18	**9)** LCD = 45 9/45, 5/45	**10)** LCD = 77 55/77, 14/77
11) LCD = 36 33/36, 4/36	**12)** LCD = 30 25/30, 14/30	**13)** LCD = 65 26/35, 20/65	**14)** LCD = 18 9/18, 16/18	**15)** LCD = 130 39/130, 90/130
16) LCD = 40 5/40, 24/40	**17)** LCD = 12 6/12, 8/12, 9/12	**18)** LCD = 60 20/60, 24/60, 25/60	**19)** LCD = 45 9/45, 10/45, 4/45	**20)** LCD = 42 36/42, 7/42, 9/42

3 Converting Improper Fractions to Whole or Mixed Numbers

Example 1

Convert 24/3 to a whole or mixed number.

Step 1 Divide the denominator (3) into the numerator (24)

$$\begin{array}{r} 8 \\ 3\overline{)\,24} \\ -24 \\ \hline 0 \end{array}$$

Since there is no remainder, 24/3 reduces to the whole number of 8.

Example 2

Convert 17/5 to a whole or mixed number.

Step 1	Divide the denominator (5) into the numerator (17)	$5\overline{)17}$ with 3 above, -15, remainder 2 Our answer is 3 remainder 2.
Step 2	Place the remainder over the original denominator (5). The number on the top of our division problem is our whole number.	3 2/5
Step 3	Check the fraction (2/5) and simplify if possible.	2/5 is fully simplified.

Example 3

Convert 18/4 to a whole or mixed number.

Step 1	Divide the denominator (4) into the numerator (18)	$4\overline{)18}$ with 4 above, -16, remainder 2 Our answer is 4 remainder 2.
Step 2	Place the remainder over the original denominator (4). The number on the top of our division problem is our whole number.	4 2/4
Step 3	Check the fraction (2/4) and simplify if possible.	2 divides evenly into both the numerator (2) and denominator (4). 2/4 ÷ 2/2 = 1/2 4 1/2 is the fully reduced mixed number.

Practice Problems

Convert each fraction to a whole or mixed number.

1) 32/9 =	**2)** 67/12 =	**3)** 116/15 =	**4)** 34/15 =	**5)** 24/12 =
6) 41/6 =	**7)** 53/7 =	**8)** 25/4 =	**9)** 127/15 =	**10)** 21/8 =
11) 16/4 =	**12)** 33/10 =	**13)** 25/9 =	**14)** 38/7 =	**15)** 99/10 =
16) 44/5 =	**17)** 53/15 =	**18)** 41/8 =	**19)** 64/9 =	**20)** 57/10 =

Answers to Practice Problems

1) 3 5/9	**2)** 5 7/12	**3)** 7 11/15	**4)** 2 4/15	**5)** 2
6) 6 5/6	**7)** 7 4/7	**8)** 6 1/4	**9)** 8 7/15	**10)** 2 5/8
11) 4	**12)** 3 3/10	**13)** 2 7/9	**14)** 5 3/7	**15)** 9 9/10
16) 8 4/5	**17)** 3 8/15	**18)** 5 1/8	**19)** 7 1/9	**20)** 5 7/10

4 Converting Improper Fractions to Whole or Mixed Numbers

Example

Convert 4 2/3 to an improper fraction.

Step 1	Multiply the whole number (4) by the denominator (3).	4 x 3 = 12
Step 2	Add the product from step 1 to the numerator (2).	12 + 2 = 14
Step 3	Place the answer from step 2 on top of the original denominator.	14/3

Practice Problems

Convert each mixed number to an improper fraction.

1) 9 1/9 =	**2)** 3 8/9 =	**3)** 8 7/12 =	**4)** 7 7/9 =	**5)** 3 11/15 =
6) 3 2/5 =	**7)** 4 2/7 =	**8)** 7 1/3 =	**9)** 5 1/7 =	**10)** 2 7/10 =
11) 3 4/5=	**12)** 4 5/7 =	**13)** 3 3/8 =	**14)** 6 1/8 =	**15)** 5 5/6 =
16) 7 4/15 =	**17)** 4 2/9 =	**18)** 9 1/6 =	**19)** 7 5/8 =	**20)** 1 5/9 =

Answers to Practice Problems

1) 82/9	**2)** 35/9	**3)** 103/12	**4)** 70/9	**5)** 56/15
6) 17/5	**7)** 30/7	**8)** 22/3	**9)** 36/7	**10)** 27/10
11) 19/5	**12)** 33/7	**13)** 27/8	**14)** 49/8	**15)** 35/6
16) 109/15	**17)** 38/9	**18)** 55/6	**19)** 61/8	**20)** 14/9

5 Adding Fractions

When adding fractions, the most important thing to remember is that the denominators **must be the same**. This is why we practiced finding a common denominator in the previous section. Also, remember to fully reduce your answers!

Example 1

Find the sum: 1/8 + 2/8 =

Step 1	Check if the denominators are the same. If they're not the same, find the common denominator.	Both denominators are 8.
Step 2	Add the numerators.	1 + 2 = 3
Step 3	Put your answer from step 2 over the common denominator.	3/8
Step 4	Check to see if your fraction can be reduced or needs to be converted from an improper fraction to a mixed number.	3/8 is a fully simplified fraction.

Example 2

Find the sum: 2/3 + 3/5 =

Step 1	Check if the denominators are the same. If they're not the same, find the common denominator.	The lowest common denominator between 3 and 5 is 15. 2/3 x 5/5 = 10/15 3/5 x 3/3 = 9/15
Step 2	Add the numerators.	10 + 9 = 19
Step 3	Put your answer from step 2 over the common denominator.	19/15
Step 4	Check to see if your fraction can be reduced or needs to be converted from an improper fraction to a mixed number.	19/15 is an improper fraction. It converts to 1 4/15.

Example 3

Find the sum: 3 3/4 + 5/6 =

Step 1	Convert any mixed numbers to improper fractions.	3 x 4 + 3 = 15. 3 ¾ as an improper fraction is 15/4.
Step 2	Check if the denominators are the same. If they're not the same, find the common denominator.	The lowest common denominator between 4 and 6 is 12. 15/4 x 3/3 = 45/12 5/6 x 2/2 = 10/12
Step 3	Add the numerators.	45 + 10 = 55
Step 4	Put your answer from step 3 over the common denominator.	55/12
Step 5	Check to see if your fraction can be reduced or needs to be converted from an improper fraction to a mixed number.	55/12 is an improper fraction. It converts to 4 7/12.

Practice Problems

Add each pair of fractions. Make sure to fully simplify your answer.

1) 1/2 + 3/15 =	**2)** 1/3+ 7/17 =	**3)** 1/3 + 3/8 =	**4)** 1/8 + 8/11 =	**5)** 4/7 +4/11 =
6) 5/6 + 1/19 =	**7)** 1/6 + 1/5 =	**8)** 1/5 + 1/2 =	**9)** 1/2+ 3/7 =	**10)** 1/3 + 11/17 =
11) 3/4 + 6/7 =	**12)** 3/7 + 14/15 =	**13)** 1/3 + 1/7 =	**14)** 4/5 + 2/13 =	**15)** 2/5 + 6/5 =
16) 1/4 + 4/7 =	**17)** 2 1/2 + 3/4 =	**18)** 5/7 + 4 3/14 =	**19)** 9 1/3 + 2 1/2 =	**20)** 5 + 4/5 =

Answers to Practice Problems

1) 7/10	**2)** 38/51	**3)** 17/24	**4)** 75/88	**5)** 72/77
6) 101/114	**7)** 11/30	**8)** 7/10	**9)** 13/14	**10)** 50/51
11) 1 17/28	**12)** 1 38/105	**13)** 10/21	**14)** 62/65	**15)** 1 3/5
16) 23/28	**17)** 3 1/4	**18)** 4 13/14	**19)** 11 5/6	**20)** 5 4/5

6 Subtracting Fractions

When subtracting fractions, the most important thing to remember is that the denominators **must be the same.** This is why we practiced finding a common denominator in the previous section. Also, remember to fully reduce your answers!

Example 1

Find the difference: 3/4 - 1/4 =

Step 1	Check if the denominators are the same. If they're not the same, then find the common denominator.	Both denominators are 4.
Step 2	Subtract the numerators.	3 - 1 = 2
Step 3	Put your answer from step 2 over the common denominator.	2/4
Step 4	Check to see if your fraction can be reduced or needs to be converted from an improper fraction to a mixed number.	2/4 = 1/2

Example 2

Find the difference: 5/8 - 1/2 =

Step 1	Check if the denominators are the same. If they're not the same, find the common denominator.	The lowest common denominator between 8 and 2 is 8. 1/2 x 4/4 = 4/8
Step 2	Subtract the numerators.	5 - 4 = 1
Step 3	Put your answer from step 2 over the common denominator.	1/8
Step 4	Check to see if your fraction can be reduced or needs to be converted from an improper fraction to a mixed number.	1/8 is a fully simplified proper fraction.

Example 3

Find the difference: 7 - 4/5 =

Step 1	Check if the denominators are the same. If they're not the same, find the common denominator.	7 is a whole number. We can think about the denominator of any whole number as 1. The lowest common denominator between 1 and 5 is 5. 7/1 x 5/5 = 35/5
Step 2	Subtract the numerators.	35 - 4 = 31
Step 3	Put your answer from step 2 over the common denominator.	31/5
Step 4	Check to see if your fraction can be reduced or needs to be converted from an improper fraction to a mixed number.	31/5 is an improper fraction. It converts to 6 1/5.

Example 4

Find the difference: 2 1/3 - 1/2 =

Step 1	Convert any mixed numbers to improper fractions.	2 x 3 + 1= 7. 2 1/3 as an improper fraction is 7/3.
Step 2	Check if the denominators are the same. If they're not the same, find the common denominator.	The lowest common denominator between 3 and 2 is 6. 7/3 x 2/2 = 14/6 1/2 x 3/3 = 3/6
Step 3	Subtract the numerators.	14 - 3 = 11
Step 4	Put your answer from step 3 over the common denominator.	11/6
Step 5	Check to see if your fraction can be reduced or needs to be converted from an improper fraction to a mixed number.	11/6 is an improper fraction. It converts to 1 5/6.

Practice Problems

1) 2 5/6 - 1/3 =	**2)** 4/5 - 3/4 =	**3)** 1/3 - 2/7 =	**4)** 8/15 - 1/4 =	**5)** 6/7 - 1/2 =
6) 3/8 - 2/11 =	**7)** 13/19 - 1/2=	**8)** 3/4 - 2/3 =	**9)** 4/7 - 1/3 =	**10)** 3 1/20 - 1/2 =
11) 1/2 - 4/11 =	**12)** 3/4 - 1/3 =	**13)** 5 1/2 - 1 4/7 =	**14)** 6/7 - 1/4 =	**15)** 3 1/4 - 1/2 =
16) 1/3 - 1/5 =	**17)** 7/11 - 2/5 =	**18)** 2/3 - 1/8 =	**19)** 1/2 - 1/7 =	**20)** 1/2 - 3/11 =

Answers to Practice Problems

1) 2 1/2	**2)** 1/20	**3)** 1/21	**4)** 17/60	**5)** 5/14
6) 17/88	**7)** 7/38	**8)** 1/12	**9)** 5/21	**10)** 2 11/20
11) 3/22	**12)** 5/12	**13)** 3 13/14	**14)** 17/28	**15)** 2 3/4
16) 2/15	**17)** 13/55	**18)** 13/24	**19)** 5/14	**20)** 5/22

7 Multiplying Fractions

Example 1

Find the product: 1/3 x 4/5 =

Step 1	Multiply the numerators.	1 x 4 = 4
Step 2	Multiply the denominators.	3 x 5 = 15
Step 3	Place the product of the numerators (4) over the product of the denominators (15).	4/15
Step 4	Check to see if the result can be simplified or reduced in any way.	4/15 is a fully simplified fraction.

Example 2

Find the product: 3/4 x 7 =

Step 1	Change any whole numbers into fractions by giving them a denominator of 1.	7 = 7/1
Step 2	Multiply the numerators.	3 x 7 = 21
Step 3	Multiply the denominators.	4 x 1 = 4
Step 4	Place the product of the numerators (21) over the product of the denominators (4).	21/4
Step 5	Check to see if the result can be simplified or reduced in any way.	2 1/4 is an improper fraction. It can be reduced to 5 1/4.

Example 3

Find the product: 2 1/2 x 5/6 =

Step 1	Change any mixed numbers into improper fractions.	2 1/2 = 5/2
Step 2	Multiply the numerators.	5 x 5 = 25
Step 3	Multiply the denominators.	2 x 6 = 12
Step 4	Place the product of the numerators (25) over the product of the denominators (12).	25/12
Step 5	Check to see if the result can be simplified or reduced in any way.	25/12 is an improper fraction. It can be reduced to 2 1/12.

Practice Problems

1) 5/7 x 4/5 =	**2)** 7/9 x 3/4 =	**3)** 1/2 x 6/7 =	**4)** 5/8 x 1/3 =	**5)** 1/9 x 1/4 =
6) 2/7 x 2/5 =	**7)** 1/9 x 1/2 =	**8)** 1/2 x 2/3 =	**9)** 6/7 x 2/3 =	**10)** 1/3 x 1/4 =
11) 5/7 x 1/2 =	**12)** 4/9 x 6/7 =	**13)** 3 x 1/6 =	**14)** 1/3 x 2 =	**15)** 2 x 1/4 =
16) 3 x 1/5 =	**17)** 2/7 x 2 1/3 =	**18)** 1 1/5 x 1/6 =	**19)** 1/9 x 1 3/7 =	**20)** 1 1/2 x 1/4 =

Answers to Practice Problems

1) 4/7	**2)** 7/12	**3)** 3/7	**4)** 5/24	**5)** 1/36
6) 4/35	**7)** 1/18	**8)** 1/3	**9)** 4/7	**10)** 1/12
11) 5/14	**12)** 8/21	**13)** 1/2	**14)** 2/3	**15)** 1/2
16) 3/5	**17)** 2/3	**18)** 1/5	**19)** 10/63	**20)** 3/8

7 Dividing Fractions

Example 1

Divide 1/4 ÷ 2/3

Step 1	Change the division sign to a multiplication sign and flip the **second** fraction.	1/4 x **3/2** **Flipping the fraction means putting the numerator in the denominator and the denominator in the numerator.**
Step 2	Multiply the fractions like normal.	1 x 3 = 3 (new numerator) 4 x 2 = 8 (new denominator) 3/8
Step 3	Check to see if the result can be reduced or simplified in any way.	3/8 is a fully reduced fraction

Example 2

Divide 5/6 ÷ 3

Step 1	Change the whole number into a fraction by adding a denominator of 1.	3 = 3/1
Step 2	Change the division sign to a multiplication sign and flip the **second** fraction.	5/6 **x 1/3**
Step 3	Multiply the fractions like normal.	5 x 1 = 5 (new numerator) 6 x 3 = 18 (new denominator) 5/18
Step 4	Check to see if the result can be reduced or simplified in any way.	5/18 is a fully reduced fraction.

Example 3

Divide 2 3/4 ÷ 1/2

Step 1	Change the mixed number into an improper fraction.	2 3/4 = 11/4
Step 2	Change the division sign to a multiplication sign and flip the **second** fraction.	11/4 **x 2/1**
Step 3	Multiply the fractions like normal.	11 x 2 = 22 (new numerator) 4 x 1 = 4 (new denominator) 22/4
Step 4	Check to see if the result can be reduced or simplified in any way.	22/4 is an improper fraction. It can be changed to a mixed number of 5 2/4. 2/4 can be simplified to 1/2, The final answer is 5 1/2.

Practice Problems

1) 2/3 ÷ 5/6 =	**2)** 1/4 ÷ 1/2 =	**3)** 1/4 ÷ 4/5 =	**4)** 1/3 ÷2/5 =	**5)** 2/3 ÷ 7/9 =
6) 2/9 ÷ 1/2 =	**7)** 2/3 ÷ 3/4 =	**8)** 1/5 ÷ 1/2 =	**9)** 2/7 ÷ 1/2 =	**10)** 1/9 ÷ 1/7 =
11) 2/3 ÷ 6 =	**12)** 1/2 ÷ 4 =	**13)** 3 ÷ 1/7 =	**14)** 1 1/2 ÷ 7/8 =	**15)** 1/7 ÷ 3 1/7 =
16) 3/5 ÷ 2 1/7 =	**17)** 2/5 ÷ 3 1/6 =	**18)** 7/9 ÷ 1 2/3 =	**19)** 3 1/8 ÷ 1 1/2 =	**20)** 4 1/2 ÷ 1 1/4 =

Answers to Practice Problems

1) 4/5	**2)** 1/2	**3)** 5/16	**4)** 5/6	**5)** 6/7
6) 4/9	**7)** 8/9	**8)** 2/5	**9)** 4/7	**10)** 7/9
11) 1/9	**12)** 1/8	**13)** 21	**14)** 1 5/21	**15)** 1/22
16) 7/25	**17)** 12/95	**18)** 7/15	**19)** 2 1/12	**20)** 3 3/5

8 Adding Decimals

Example

Find the sum. 28.4 + .04 + 36 + 1.345

Step 1	Add a decimal at the end of any whole numbers so that every number in the problem has a decimal.	36 = 36.
Step 2	Rewrite the problem vertically so that the numbers line up **at the decimal point**.	28.4 + .04 36. 1.345
Step 3	Add the columns vertically. Make sure to bring the decimal point directly down below the answer line.	^{1} 28.4 + .04 36. 1.345 65.785

9 Subtracting Decimals

Example

Find the difference. 47 - 36.34

Step 1	Add a decimal at the end of any whole number so that every number in the problem has a decimal.	47 = 47.
Step 2	Add enough zeros **after** the decimal so that each number has the same number of digits **after** the decimal. The number of digits before the decimal do not need to be the same.	47. = 47.00
Step 3	Rewrite the problem vertically so that the numbers line up **at the decimal point**.	47.00 - 36.34
Step 4	Subtract the columns vertically. Make sure to bring the decimal point down directly below the answer line.	$^{6\ 9\ 10}$ 4~~7~~.~~0~~~~0~~ - 36.34 10.66

Practice Problems

1) 36.2 + 14 + .09 =	**2)** 1.004 + 2.4 =	**3)** 1.23 + 2.24 + 3 =	**4)** 6.0003 + 2.01 =	**5)** 1.03 + 33.42 =
6) 7 + .03 + .123 =	**7)** 14 + 26.3 + .34 =	**8)** 2.23 + 3.345 + 6 =	**9)** 3.32 + 1.43 =	**10)** 9.32 + .1 + .339 =
11) 8 - 6.45 =	**12)** 23.1 - 2.33 =	**13)** 21 - 8.0083 =	**14)** 3.42 - 2 =	**15)** 19 - 3.657 =
16) 16.33 - 2.34443 =	**17)** 5.4 - .9332	**18)** 18 - .003 =	**19)** 3.001 - 1.23 =	**20)** 7.61 - 4.32 =

Answers to Practice Problems

1) 50.29	**2)** 3.404	**3)** 6.47	**4)** 8.0103	**5)** 34.45
6) 7.153	**7)** 40.64	**8)** 11.575	**9)** 4.75	**10)** 9.759
11) 1.55	**12)** 20.77	**13)** 12.9917	**14)** 1.42	**15)** 15.343
16) 13.98557	**17)** 4.4668	**18)** 17.997	**19)** 1.771	**20)** 3.29

10 Multiplying Decimals

Example 1

Find the product. 32.4 x 6.03 =

Step 1 Rewrite the problem vertically and aligned with the last number (like you would if there **were not** any decimals).

$$\begin{array}{r} 32.4 \\ \times\ 6.03 \\ \hline \end{array}$$

Step 2 Multiply as normal, ignoring the decimals.

$$\begin{array}{r} {\scriptstyle 1\ \ \ \ } \\ 32.4 \\ \times\ 6.03 \\ \hline 972 \end{array} \qquad \begin{array}{r} 32.4 \\ \times\ 6.03 \\ \hline 972 \\ 0000 \end{array} \qquad \begin{array}{r} {\scriptstyle 1\ 2\ \ \ } \\ 32.4 \\ \times\ 6.03 \\ \hline 972 \\ 0000 \\ 19400 \end{array} \qquad \begin{array}{r} 32.4 \\ \times\ 6.03 \\ \hline 972 \\ +\ 0000 \\ \ 19400 \\ \hline 195372 \end{array}$$

Step 3 Count the number of digits to the in the problem to the **right** of the decimal.

32.**4** x 6.**03**
There are 3 total digits to the right of the decimal.

Step 4 Take your answer from step 2 and move the decimal to the left 3 (the number from step 3) digits.

195372 to 195.372
The decimal is moved 3 spaces to the left.

Example 2

Find 6% of 34.

Step 1 Change the percentage to a decimal by moving the decimal two digits to the left.

6% = .06

Step 2 Rewrite the problem vertically, using the decimal version of the percentage, aligned with the last number (like you would if there **were not** any decimals).

$$\begin{array}{r} .06 \\ \times\ 34 \\ \hline \end{array}$$

Step 3 Multiply as normal, ignoring the decimals.

$$\begin{array}{r} {\scriptstyle 2} \\ .06 \\ \times\ 34 \\ \hline 24 \end{array} \qquad \begin{array}{r} {\scriptstyle 1} \\ .06 \\ \times\ 34 \\ \hline 24 \\ 180 \end{array} \qquad \begin{array}{r} .06 \\ \times\ 34 \\ \hline {\scriptstyle 1}24 \\ +\ 180 \\ \hline 204 \end{array}$$

Step 4 Count the number of digits to the in the problem to the **right** of the decimal.

.06 x 34
There are 2 total digits
to the right of the decimal.

Step 5 Take your answer from step 2 and move the decimal to the left 3 (the number from step 3) digits.

204 to 2.04
The decimal is moved
2 spaces to the left.

11 Dividing Decimals

Example 1

Divide 3.75 ÷ 3 =

Step 1 Set up the problem as a regular long division problem.

$$3\overline{)\,3.75}$$

Step 2 Bring up the decimal point for the answer directly above the decimal point in the problem.

$$3\overline{)\,3.75}$$

Step 3 Divide as normal.

$$\begin{array}{r} 1. \\ 3\overline{)\,3.75} \\ -\,3\downarrow \\ \hline 07 \end{array} \qquad \begin{array}{r} 1.2 \\ 3\overline{)\,3.75} \\ -\,3\downarrow \\ \hline 07 \\ -\,6\downarrow \\ \hline 15 \end{array} \qquad \begin{array}{r} 1.25 \\ 3\overline{)\,3.75} \\ -\,3\downarrow \\ \hline 07 \\ -\,6\downarrow \\ \hline 15 \\ -\,15 \\ \hline 0 \end{array}$$

The answer is 1.25.

Example 2

Divide 6.2 ÷ .04 =

Step 1 Set up the problem as a regular long division problem.

$$.04\overline{)\,6.2}$$

Step 2 The number on the outside needs to become a whole number. To do that, we need to move the decimal point to the right however many spaces to make it a whole number.

$$04.\overline{)\,6.2}$$

Since .04 has two numbers to the right of the decimal, we need to move the decimal 2 spaces to the right. .04 becomes 04. or, more simply, 4.

Step 3 Move the decimal of the inside number **the same number of spaces to the right** as the number on the outside.

Note 1: It is okay if the inside number is still a decimal.

Note 2: If there aren't enough digits in the number (or it is a whole number), we will need to add 0s to the end of the number.

$$4\overline{)\,620.}$$

6.2 becomes 620. Since 6.2 only has 1 number to the right of the decimal, we added a 0 to fill the extra space (see Note 2 on the left).

Step 4 Bring up the decimal point for the answer directly above the decimal point in the problem.

In this problem, there is no longer a decimal inside the division sign, so there is no decimal to bring up. But, don't forget to check for one when dividing with decimals!

Step 5 Divide as normal.

$$\begin{array}{r} 1 \\ 4\overline{)\,620} \\ -\,4\downarrow \\ \hline 22 \end{array} \qquad \begin{array}{r} 15 \\ 4\overline{)\,620} \\ -\,4\downarrow \\ \hline 4\overline{)\,22}\downarrow \\ -\,20 \\ \hline 20 \end{array} \qquad \begin{array}{r} 155 \\ 4\overline{)\,620} \\ -\,4\downarrow \\ \hline 4\overline{)\,22}\downarrow \\ -\,20 \\ \hline 4\overline{)\,20} \\ -\,20 \\ \hline 0 \end{array}$$

The answer is 155.

Practice Problems

1) .26 x 4.4 =	**2)** .68 x 5.6 =	**3)** .84 x 8.2 =	**4)** 6.7 x .42 =	**5)** 8.6 x .33 =
6) 36.8 x 6.3 =	**7)** .806 x 7.6 =	**8)** 6.39 x 9.4 =	**9)** 43% of 38	**10)** 84% of 165
11) 8.72 ÷ 5	**12)** 1.4 ÷ 4 =	**9)** 1.12 ÷ 7 =	**14)** 7.82 ÷ 2 =	**15)** 155.2 ÷ 2 =
16) 499.29 ÷ 5.1 =	**17)** 467.22 ÷ 7.8 =	**18)** .7959 ÷ .21 =	**19)** 5.7288 ÷ .93 =	**20)** 2.0853 ÷ .21 =

Answers to Practice Problems

1) 1.144	**2)** 3.808	**3)** 6.888	**4)** 2.814	**5)** 2.838
6) 231.84	**7)** 6.1256	**8)** 60.066	**9)** 16.34	**10)** 138.6
11) 1.744	**12)** .35	**13)** .16	**14)** 3.91	**15)** 77.6
16) 97.9	**17)** 59.9	**18)** 3.79	**19)** 6.16	**20)** 9.93

Practice Test

This is a practice numerical computation test. You **should not** use a calculator for this. You may use scratch paper. You have 35 minutes to complete this practice test.

1) 24 + 36 - 4	A) 50	B) 66	C) 56	D) 60	E) 54
2) 2.9 x 6.3	A) 18.27	B) 182.7	C) 19.26	D) 18.26	E) 1.82
3) 1/3 x 3 3/4	A) 1 1/4	B) 2 1/3	C) 3 1/2	D) 3 2/3	E) 2 3/4
4) 5/12 + 2/3 + 5/6	A) 1 5/6	B) 1 11/12	C) 2 11/12	D) 2 5/6	E) 2
5) 1/2 + 3/4 + 4/5 + 9/10	A) 19/20	B) 2 3/4	C) 3 4/5	D) 2 19/20	E) 2 4/5
6) 3 7/8 - 2 1/4	A) 1 3/5	B) 1 3/4	C) 1 1/2	D) 1	E) 1 5/8
7) 18% of 50	A) 10	B) 9	C) 5	D) 20	E) 15
8) 6.4 ÷ 4	A) 16	B) .16	C) 1.6	D) 4.6	E) .46
9) 3 x (6 + 2)	A) 20	B) 24	C) 12	D) 16	E) 36
10) 4/7 ÷ 3/8	A) 3/14	B) 7/15	C) 2 2/3	D) 1 11/21	E) 1 10/21
11) 2.34 + 8.7	A) 10.03	B) 11.54	C) 16.9	D) 11.04	E) 6.4
12) 5.35 ÷ .5	A) 1.05	B) 25.5	C) 15.3	D) 10.7	E) 10.54

13) 2.36 ÷ .4	A) 8.6	B) .563	C) 5.9	D) .59	E) 9.44
14) 28% of 96	A) 26.88	B) 34.23	C) 18.74	D) 14.23	E) 7.93
15) 38.43 - 7.62	A) 30.81	B) 31.21	C) 46.05	D) 18.34	E) 21.98
16) 10.07 - 6.89	A) 16.96	B) 4.22	C) 3.64	D) 4.76	E) 3.18
17) 6 1/2 x 3 2/3	A) 18 1/3	B) 6 2/3	C) 23 5/6	D) 23 1/3	E) 18 4/5
18) 4/9 + 3/7	A) 7/9	B) 55/63	C) 57/63	D) 18/19	E) 1/2
19) 5 4/5 ÷ 8	A) 46 2/5	B) 37/40	C) 5/8	D) 23/40	E) 29/40
20) 6.4 ÷ 4	A) 16	B) .16	C) 1.6	D) 4.6	E) .46
21) 5 3/4 + 6 7/8	A) 11 7/8	B) 11 5/6	C) 12 2/3	D) 12 5/8	E) 12 1/2
22) 3 ÷ (1/2 + 3/4)	A) 2 2/5	B) 4 1/2	C) 6 3/4	D) 3 2/5	E) 3 1/2
23) 5/12 ÷ 6/7	A) 24/25	B) 3/7	C) 30/41	D) 3/4	E) 35/72
24) 8.04 - 2.67	A) 10.71	B) 6.43	C) 6.33	D) 5.43	E) 5.37
25) 5% of 64	A) 32	B) .32	C) 3.2	D) 6.4	E) 1.5
26) 8/9 - 2/3	A) 1/9	B) 2/9	C) 1/3	D) 2/3	E) 4/9
27) 6 4/5 - 3	A) 3 1/2	B) 3 4/5	C) 3 1/3	D) 3 1/5	E) 3 2/5
28) 5.5 x 2.35	A) 12.925	B) 1.2925	C) 129.25	D) 11.342	E) 1.1342
29) 4.2 ÷ .007	A) 6	B) 7	C) 60	D) 600	E) 42
30) 84 + 36 -17	A) 110	B) 102	C) 103	D) 111	E) 107

Answers to Practice Test

1) C) 56	**2)** A) 18.27	**3)** A) 1 1/4	**4)** B) 1 11/12	**5)** D) 2 19/20
6) E) 1 5/8	**7)** B) 9	**8)** C) 1.6	**9)** B) 24	**10)** D) 1 11/21
11) D) 11.04	**12)** D) 10.7	**13)** C) 5.9	**14)** A) 26.88	**15)** A) 30.81
16) E) 3.18	**17)** C) 23 5/6	**18)** E) 29/40	**19)** B) 55/63	**20)** E) 16/21
21) D) 12 5/8	**22)** A) 2 2/5	**23)** E) 35/72	**24)** E) 5.37	**25)** C) 3.2
26) B) 2/9	**27)** B) 3 4/5	**28)** A) 12.925	**29)** D) 600	**30)** C) 103

Numerical Reasoning

Overview

The Numerical Reasoning test assesses your ability to recognize and identify patterns in a series of numbers. To correctly answer the questions on this test, you first need to identify the pattern and then apply it to calculate the next number in the series.

Test-Taking Tips

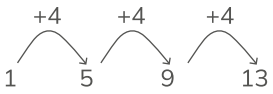
- Read the entire series of numbers. Some patterns will be identifiable from the first two numbers, but not all. Make sure to check that your identified pattern works with all the numbers in the series.
- Try to identify the operation(s) for the pattern before you begin solving.
    - If the numbers are increasing, the pattern will likely use addition or multiplication.
    - If the numbers are decreasing, the pattern will likely use subtraction or division.
    - If the numbers are 'both' increasing and decreasing, the pattern will likely use a combination of addition or multiplication and subtraction or division.
    - If you can't identify a common operation for the pattern, try thinking outside the box and identify what else the numbers have in common.
- If the pattern starts out with very large numbers and decreases, work backwards and try to identify the pattern using the smaller numbers at the end first.
- Use your scratch paper.
- Check that your answer makes sense. For example, if the pattern is increasing, and your answer is smaller, you likely made a mistake.
- If your answer isn't a choice, double-check your calculations and that your pattern works for all the given numbers.
- If you are stuck on a question, skip it and come back to it at the end. Make sure not to spend too much time on any one question.
- Remember, fractions in the series will be fully simplified. It may help you identify the pattern to think about the unsimplified fraction.

Examples & Practice Problems

1 Using One Operation

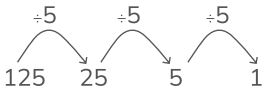
Example 1

Find the next number in the series: 1, 5, 9, 13...

Step 1	Identify if the series is increasing, decreasing, both, or neither.	The series is increasing, so it will likely use addition or multiplication.
Step 2	Use your scratch paper to identify how the series is changing.	 We are adding 4 to each number.
Step 3	Use the pattern to calculate the next number in the series.	13 + 4 = 17. The next number in the series is 17.

Example 2

Find the next number in the series: 125, 25, 5, 1...

Step 1	Identify if the series is increasing, decreasing, both, or neither.	The series is decreasing, so it will likely use subtraction or division.
Step 2	Use your scratch paper to identify how the series is changing.	 We are dividing each number by 5.
Step 3	Use the pattern to calculate the next number in the series.	1 ÷ 5 = 1/5. The next number in the series is 1/5.

Practice Problems

Find the next number in the series.

1) 3, 6, 12, 24...	**2)** 1/2, 1, 1 1/2, 2...	**3)** 87, 77, 67, 57...
4) 54, 18, 6, 3...	**5)** 1/2, 1/4, 1/8, 1/16...	**6)** 7/8, 3/4, 5/8, 1/2...
7) .2, .25, .3, .35...	**8)** 3, 9, 27, 81...	**9)** 10, 14, 18, 22...

Answers to Practice Problems

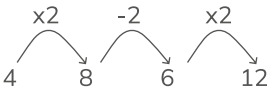
Method for answer solution in parenthesis

1) 48 (x 2)	**2)** 2 1/2 (+ 1/2)	**3)** 47 (- 10)
4) 1 (÷ 3)	**5)** 1/32 (x 2)	**6)** 3/8 (- 1/8)
7) .4 (+ .05)	**8)** 243 (x 3)	**9)** 26 (+ 4)

2 Using 2 Operations

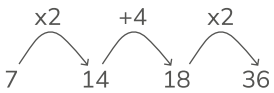
Example 1

Find the next number in the series: 4, 8, 6, 12...

Step 1	Identify if the series is increasing, decreasing, both, or neither.	The series is both increasing and decreasing, so it will likely use addition or multiplication **and** subtraction or division.
Step 2	Use your scratch paper to identify how the series is changing.	 We are alternating between multiplying by 2 and subtracting 2.
Step 3	Use the pattern to calculate the next number in the series.	12 - 2 = 10. The next number in the series is 10.

Example 2

Find the next number in the series: 7, 14, 18, 36...

Step 1	Identify if the series is increasing, decreasing, both, or neither.	The series is increasing, so it will likely use addition or multiplication.
Step 2	Use your scratch paper to identify how the series is changing.	 This pattern uses both addition **and** multiplication.
Step 3	Use the pattern to calculate the next number in the series.	36 + 4 = 40. The next number in the series is 40.

Practice Problems

Find the next number in the series.

1) 5, 10, 7, 12, 9...	**2)** 1/2, 1, 1/3, 2/3...	**3)** 2, 4, 6, 12, 14...
4) 500, 100, 50, 10...	**5)** 6, 4, 7, 5...	**6)** 36, 31, 93, 88...
7) 4, 400, 40, 4000	**8)** 17, 23, 20, 26...	**9)** 5, 10, 30, 60...

Answers to Practice Problems

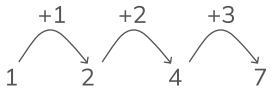
Method for answer solution in parenthesis

1) 14 (+ 5, - 3)	**2)** 1/9 (x 2, ÷ 3)	**3)** 28 (x 2, + 2)
4) 5 (÷ 5, ÷ 2)	**5)** 8 (- 2, +3)	**6)** 264 (- 5, x 3)
7) 400 (x 100, ÷ 10)	**8)** 23 (+6, -3)	**9)** 180 (x 2, x 3)

3 Other Types of Patterns

Example 1

Find the next number in the series: 1, 2, 4, 7...

Step 1	Identify if the series is increasing, decreasing, both, or neither.	The series is increasing, so it will likely use addition or multiplication.
Step 2	Use your scratch paper to identify how the series is changing.	 In this series, the number you are adding increases by 1 each time.
Step 3	Use the pattern to calculate the next number in the series.	7 + 4 = 11. The next number in the series is 11.

Example 2

Find the next number in the series: 1, 4, 9, 16...

Step 1	Identify if the series is increasing, decreasing, both, or neither.	The series is increasing, so it will likely use addition or multiplication.
Step 2	Use your scratch paper to identify how the series is changing.	1^2 2^2 3^2 4^2 ↑ ↑ ↑ ↑ 1 4 9 16 In this series, the number we are squaring is increasing by 1 each time.
Step 3	Use the pattern to calculate the next number in the series.	$5^2 = 25$. The next number in the series is 25

Practice Problems

1) 0, 0, 2, 6, 12...	**2)** 5, 10, 30, 120...	**3)** 3125, 625, 125, 25...
6) 7, 8, 6, 8, 4, 7, 1...	**7)** 2040, 408, 102, 34...	**8)** 57, 56, 54, 51...

Answers to Practice Problems

Method for answer solution in parenthesis

1) 20 (+0, +2, +4, +6, +8...)	**2)** 600 (x2, x3, x4, x5...)	**3)** 5 (5^5, 5^4, 5^3, 5^2, 5^1...)
6) 5 (+1, -2, +2, -4, +3, -6, +4...)	**7)** 17 (÷5, ÷4, ÷3, ÷2...)	**8)** 47 (-1, -2, -3, -4...)

Practice Test

This is a practice numerical reasoning test. You **should not** use a calculator for this. You may use scratch paper. You have 15 minutes to complete this practice test.

1) 4, 6, 8, 10, 12	A) 16	B) 20	C) 18	D) 14	E) 15
2) 21, 42, 63, 84, 105	A) 112	B) 126	C) 116	D) 134	E) 110
3) 2, 11, 9, 18, 16	A) 32	B) 14	C) 25	D) 30	E) 26
4) 2, 4, 8, 16, 32	A) 48	B) 64	C) 126	D) 128	E) 50
5) 7, 12, 24, 29, 58	A) 63	B) 116	C) 60	D) 82	E) 31
6) 1/2, 2/3, 3/4, 4/5, 5/6	A) 7/8	B) 5/7	C) 5/8	D) 9/10	E) 6/7
7) 5, 5, 10, 30, 120	A) 240	B) 180	C) 720	D) 600	E) 560
8) 1458, 162, 18, 2, 2/9	A) 1/9	B) 2/81	C) 1/3	D) 1/81	E) 4/9
9) 4, 5, 10, 11, 22	A) 23	B) 44	C) 24	D) 21	E) 30
10) 70, 7, .7, .07, .007	A) .00007	B) 700	C) 7/10	D) 7/100	E) .0007
11) 910, 900, 450, 440, 220	A) 110	B) 200	C) 210	D) 230	E) 100
12) 1000, 200, 40, 8, 1.6	A) 3.2	B) .16	C) .32	D) .8	E) .4
13) 7, 8, 16, 19, 76	A) 77	B) 81	C) 152	D) 228	E) 75
14) 1/2 1/3 2/9 4/27 8/81	A) 16/243	B) 9/162	C) 9/243	D) 16/162	E) 8/243

15) 14, 15, 15, 16, 16	A) 16	B) 15	C) 14	D) 17	E) 18
16) 84, 82, 78, 72, 64	A) 52	B) 56	C) 55	D) 53	E) 54
17) 972, 324, 108, 36, 12	A) 3	B) 6	C) 8	D) 4	E) 2
18) 1, 6, 36, 216, 1296	A) 7776	B) 3888	C) 6480	D) 5184	E) 9072
19) 1/2, 1/4, 1/8, 1/16, 1/32	A) 1/81	B) 3/64	C) 1/64	D) 2/81	E) 1/9
20) 3, 15, 14, 70, 69	A) 276	B) 68	C) 414	D) 72	E) 345

Answers to Practice Test

1) D) 14	**2)** B) 126	**3)** C) 25	**4)** B) 64	**5)** A) 63
6) E) 6/7	**7)** D) 600	**8)** B) 2/81	**9)** A) 23	**10)** E) .0007
11) C) 210	**12)** C) .32	**13)** B) 81	**14)** A) 16/243	**15)** D) 17
16) E) 54	**17)** D) 4	**18)** A) 7776	**19)** C) 1/64	**20)** E) 345

Paper Folding

Overview

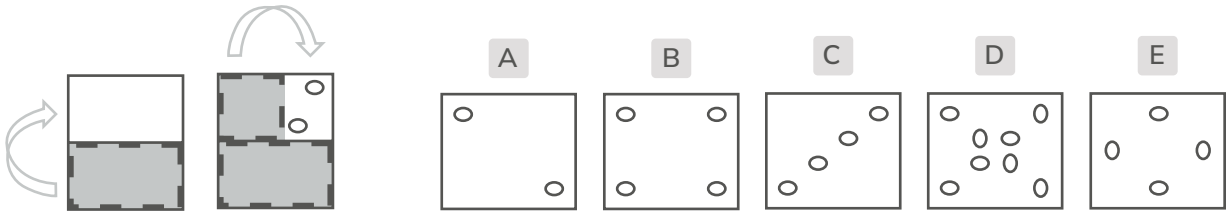
The paper folding test is designed to identify how well you can visualize in a three-dimensional space. Each question on this test will show you a folded square of paper with one or more holes punched through it. You will need to visualize what the paper will look like once it is unfolded. Then, you will have to determine how many holes there will be and where they will be located on the fully unfolded paper. You **will not** have any scratch paper for this test.

Test-Taking Tips

- Pay attention to how many times the paper is folded and how many holes have been punched. Remember, each fold effectively **doubles** the sheets of paper being punched through.
- Read each question carefully. Make sure to note all the folds and all the holes that have been punched.
- Do your best to imagine the paper in three dimensions and visualize where the holes would be on an unfolded paper. Remember, you cannot actually turn or fold a paper on the test.
- Once you've found the answer that you think is closest, check your answer by trying to mentally refold the paper to see if it matches the folded paper in the question.
- For some problems, the paper may need to be rotated after it has been unfolded in order to find the correct answer.
- Create your own practice problems by taking a stack of post-its, folding them in various ways and then punching holes. Try to predict what the post-it will look like when you unfold the paper.
- The more you practice, the better you will be able to visualize on the test!

Examples

Example 1



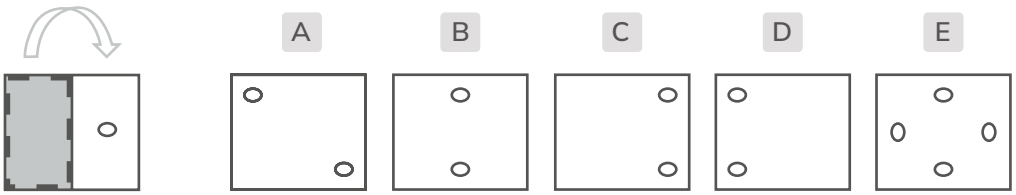
Step 1 Note that the paper is folded two times, once from bottom to top and then from left to right. This creates **four** layers of paper to punch through.

Step 2 Two holes are punched in the folded paper, one in the top right corner and one in the bottom left corner of the folded paper.

Step 3 When unfolded, the hole that was punched in the top right corner will result in four holes, one in each corner. The hole that was punched in the bottom left of the folded paper will result in four holes, creating a smaller square around the middle of the paper.

Step 4 D is the correct answer, as it shows 8 total holes, one in each corner, and 4 in a square around the middle.

Example 2



Step 1 Note that the paper is folded one time from the left to the right. This creates **two** layers of paper to punch through.

Step 2 One hole is punched in the folded paper, in the center of the folded side.

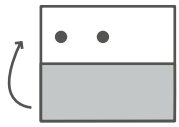
Step 3 When unfolded, the hole that was punched creates two holes, one on each side of the center line.

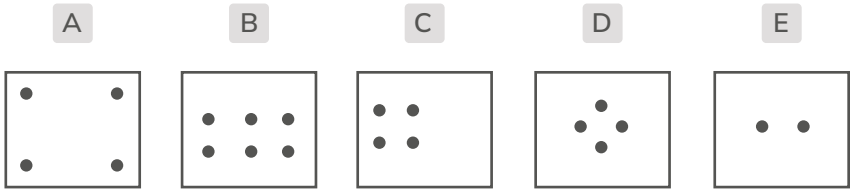
Step 4 B is the correct answer. To obtain B, we need to rotate our unfolded paper so that the punched holes move from the center of the sides to the center of the top and bottom.

Practice Test

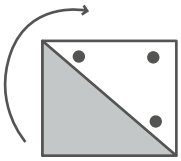
This is a practice paper folding test. You **cannot** use scratch paper or fold/manipulate paper in any way during this test. You have 26 minutes to complete this practice test.

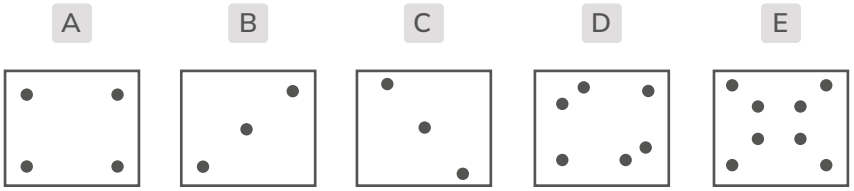
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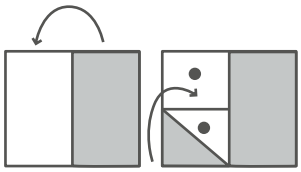


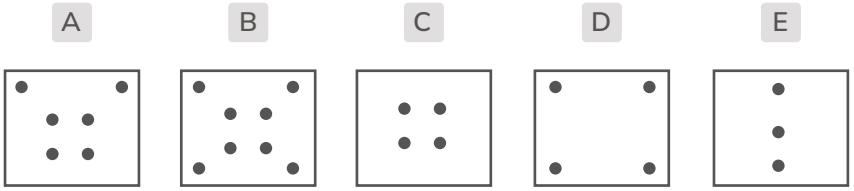
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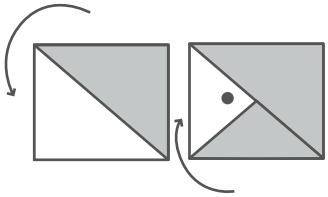


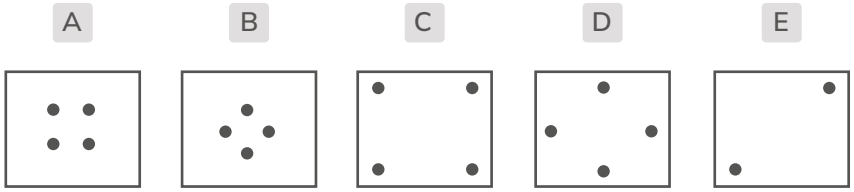
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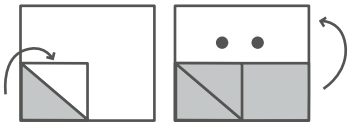


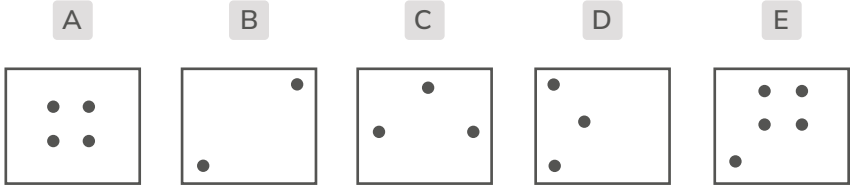
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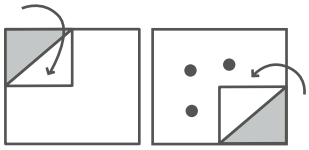


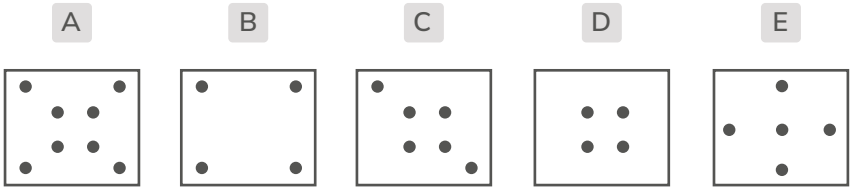
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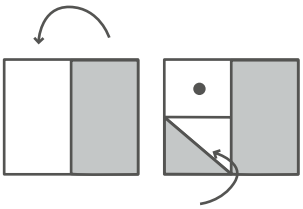


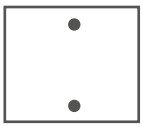
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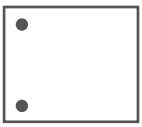


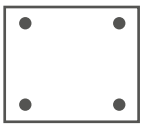


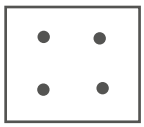
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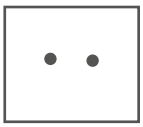


A 

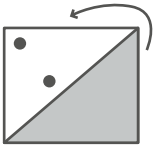
B 

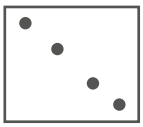
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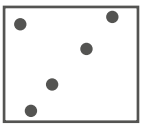
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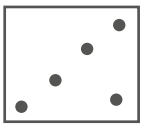
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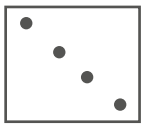
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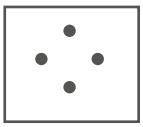


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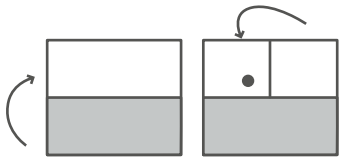
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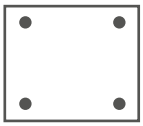
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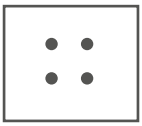
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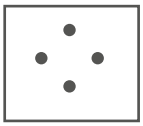
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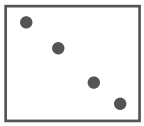
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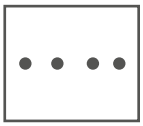


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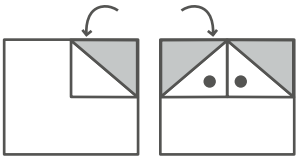
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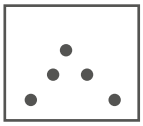
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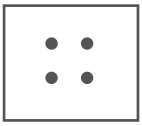
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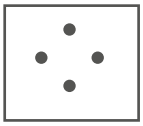
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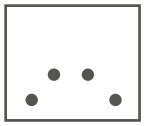
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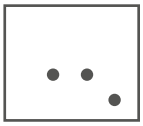


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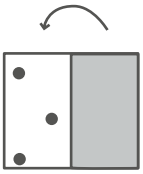
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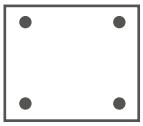
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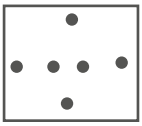
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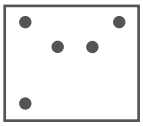
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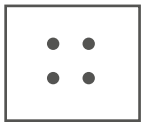
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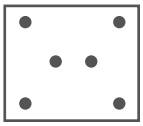


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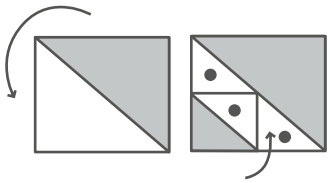
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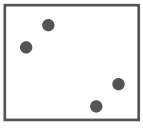
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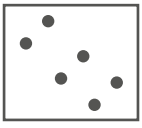
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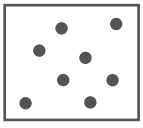
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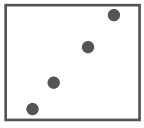
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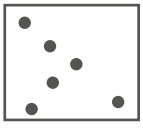


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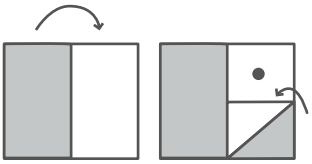
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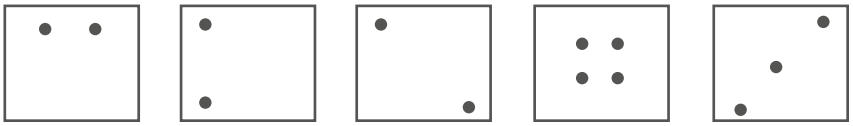
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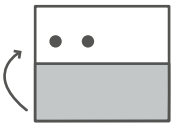
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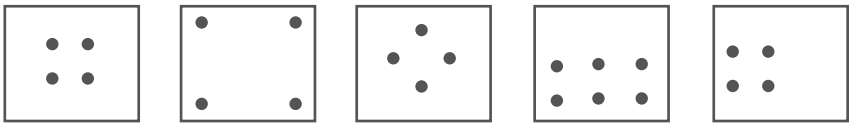
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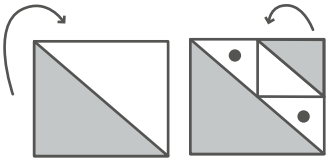
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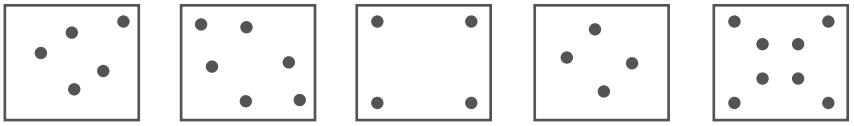
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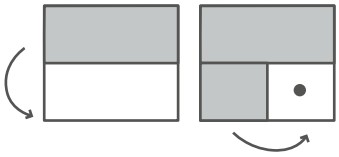
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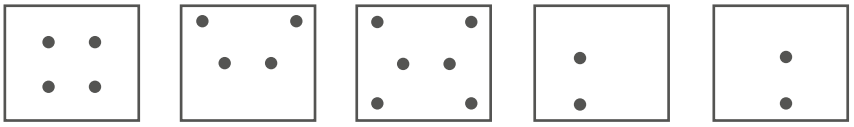
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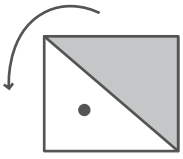
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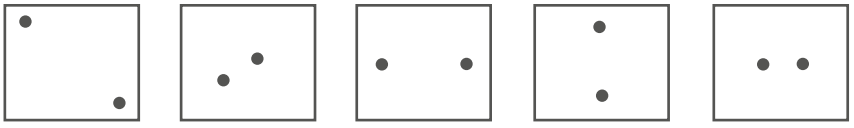
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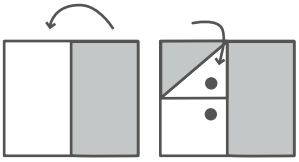
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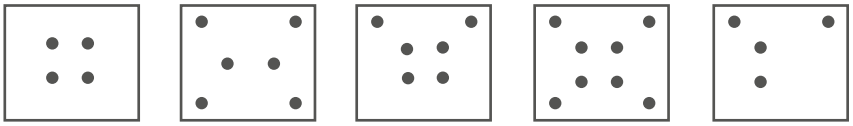
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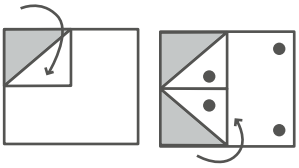
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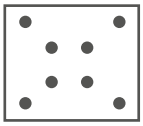


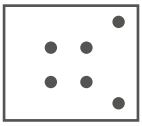
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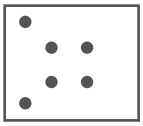


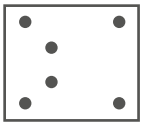
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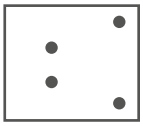


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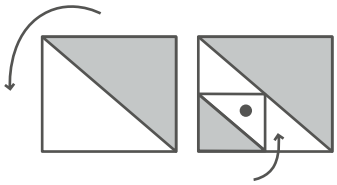
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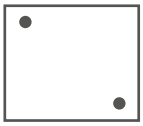
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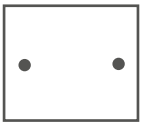
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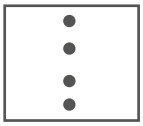
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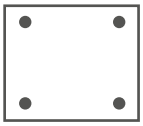
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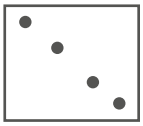


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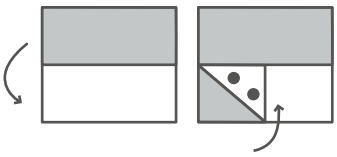
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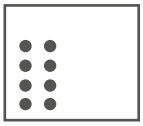
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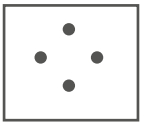
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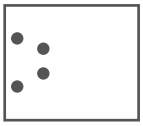
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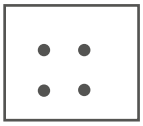
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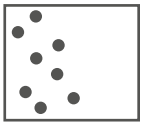


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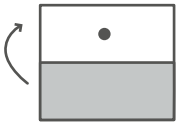
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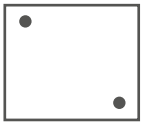
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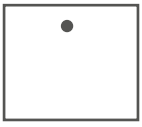
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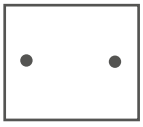
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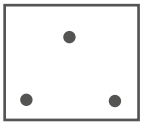
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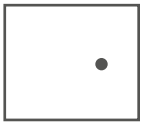


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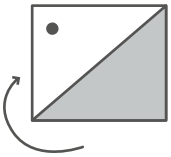
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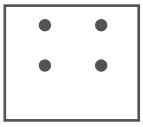
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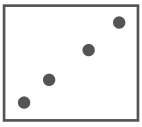
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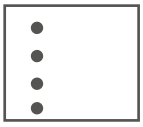
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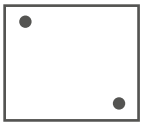
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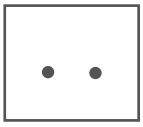


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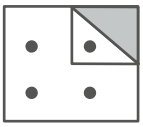
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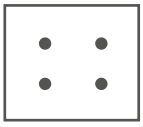
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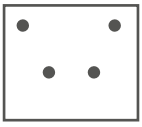
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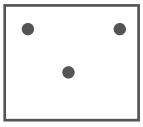
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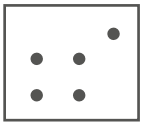
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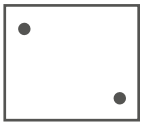


A 

B 

C 

D 

E 

Answers to Practice Test

1) C	**2)** D	**3)** A	**4)** B	**5)** E	**6)** C
7) E	**8)** A	**9)** B	**10)** D	**11)** E	**12)** C
13) A	**14)** E	**15)** D	**16)** A	**17)** B	**18)** C
19) D	**20)** E	**21)** E	**22)** C	**23)** D	**24)** D

Reading Comprehension

Overview

This test assesses how well you comprehend information from texts that you have read. The test consists of a series of short paragraphs, each followed by a few comprehension questions about the paragraph.

Test-Taking Tips

- Read the questions first, then go back and read the paragraph. This allows you to search for the needed information while you are reading, ultimately saving you time in the end.
- Make sure to read the text and the passage thoroughly. You will waste more time skimming and having to continually reread the passage, rather than reading it closely the first time.
- Make sure to read all the answers before selecting the best answer. Some answers may be similar, but there will always be one **best** answer.
- Look for keywords in the text and the question. This will help you identify where in the passage the answer may be located.
- Remember, this is a comprehension test, not a memory test. You can refer back to the text to answer the questions so don't try to remember the details.

Practice Test

This is a practice reading test. You will have 13 minutes to complete this test.

1 Scenario A

Several types of girders are used in steel fabrication. The most common types are the inverted girder, the beam-like girder, and the box girder. Girders are built-up beams and are fabricated in many different sizes from plate stock. The inverted girder is used mainly for supporting long span roofs, such as warehouses and factories. The beam-like girder is used for highway and railroad bridges, which are sometimes as long as 120 feet. The box girder is also used in highway and railroad applications, with the ability to hold heavy loads. Girders usually have connection stiffeners on both sides and bearing stiffeners at stress points. Shoe plates are used at the stress points where the girder will rest on the piers.

---

1. For steel fabrication, one of the most common type of girder is called

A) box
B) in-line
C) round
D) straight

---

2. An inverted girder would be used in the construction of a

A) swimming pool
B) factory building
C) storage shed
D) garage

---

3. From the information above, the type of girder chosen depends on

A) preference
B) location
C) the type of building
D) tools for installation

---

2 Scenario B

The size of screwdrivers is measured by the length of the blade and not the overall length of the screwdriver. The diameter of the blade increases as the length of the blade increases. Also, the tip is wider and thicker as the length of the blade increases. Finally, the slots in the screw heads become wider as the screw size increases. As a result, the size of the screwdriver can match the size of the screw. A screwdriver with the largest tip should be used for the screw. If the screwdriver is too small for the size of the screw, it will be difficult to turn the screw or the screw will be damaged. The screwdriver blade should always be kept perpendicular to the screw head. Also, the screwdriver must be pushed as it is turned to hold it in the screw slot. Hammering on the screwdriver should be avoided. Hammering on the screwdriver usually means the wrong tool is being used, rather than a chisel, punch, or nail set.

---

4. As the length of the blade increases, the width of the tip is

A) narrower and thicker
B) wider and thinner
C) remains constant
D) wider and thicker

---

5. When fastening a screw, the screwdriver should be

A) as small as possible
B) at an angle to the crew head
C) hit with a hammer
D) pushed as it is turned

---

6. As the length of the blade increases, the diameter of the blade

A) decreases
B) increases
C) remains the same
D) depends on the material

---

3 Scenario C

Pliers are holding tools which increase the usefulness of your hands by increasing gripping power or holding hot or cold objects. Plier sizes are determined by the length of the pliers. The larger the size, the more powerful the pliers. Adjustable joint pliers can be set so that the jaws pivot, changing the maximum opening of the jaws. This allows the jaw opening to match the size of the object to be gripped. Linemen's side cutting pliers are useful for bending or cutting wire. Diagonal cutting pliers have sloping cutting edges but have no jaws for gripping. They are useful in mechanical work requiring cotter pins. Vise grip pliers are a multi-purpose tool that are both pliers and a clamp or vise. When the handles are closed together the jaws close on the work.

7. For work on large, heavy duty bolts, the pliers should be

A) smaller
B) wide
C) larger
D) size does not matter

8. Lineman's side cutting pliers would be useful for a/an

A) ironworker
B) sheet metal worker
C) pipefitter
D) electrician

9. Vise grips are multi-purpose tools that work as

A) pliers
B) clamps
C) vises
D) both pliers and clamps or vises

4 Scenario D

Slope or slant roofs are commonly used for homes, small commercial buildings, and churches. The distance from one roof eave to the opposite roof eave is called the span. Half the distance of the span is the run. There are many designs for steep slope roofs. For example, a gable roof is shaped like an inverted V. It forms a ridge where the two sloped sections join. A hip roof slopes in four directions from a common ridge. A simple hip roof joins in a peak. A gambrel roof slopes in two directions from a common ridge with a break in the top sloping fields. Other types of steep sloped roofs are shed roofs and mansard roofs.

---

10. The slant of a roof can be described with the term

A) slope
B) rise
C) degree
D) hip

---

11. The distance from one roof eave to the next is called

A) span
B) pitch
C) hip
D) run

---

12. Half the span of a roof is called the

A) rise
B) run
C) pitch
D) slope

---

5 Scenario E

The shapes and dimensions of pipe threads have been established according to nationally accepted standards. A thread on the inside of a pipe is an internal thread, while the thread on the outside of the pipe is an external thread. When threads are properly cut there will be little difficulty in making a fluid tight joint. A die is used to cut external threads. A tap is used to cut internal threads. In a full width die, the width of the die must equal the required length of the thread. The other type of die is the receding die. The width of the receding die is less than the length of the thread to be cut, so the die travels along the pipe. Full width dies are used for smaller pipe sizes while receding dies are used for larger size pipes. A group of die stocks are also called non-ratcheting. Because they do not move, they must be hand operated or moved with power equipment. The pipe itself is moved, but the die remains stationary.

13. Internal threads are cut using a

A) die.
B) tap.
C) joint.
D) pipe.

14. A non-ratcheting stocks move by

A) hand
B) automatically
C) power equipment or automatically
D) power equipment or by hand

15. The dimensions of pipe threads are determined

A) by the manufacturer
B) according to state-wide accepted standards
C) according to nationally accepted standards
D) according to globally accepted standards

6 Scenario F

There are a variety of drills and drill bits available for industrial applications. The most common method of designating sizes is by their diameter in fractions of an inch. One end of the drill bit is the cutting edge and the opposite end is called the shank. There are variations of the shank, depending upon how much grip is needed to complete the work. Types of shanks are called straight, round end, hexagon, or flats ground. There is also a tapered shank bit, which is used in stationary drill presses and some heavy duty portable tools. Tapered square shanks are used with a bit brace drill. A twist drill is used for drilling holes in metal. Several types of bits are available for wood boring. Wood bits for power tools are flat or multi-spur. A square taper shank is for use with a bit brace. Other square taper shank bits are the solid center auger bit, alternate round shank, double twist auger bit, or single twist auger bit. A solid center is stiff and strong for general wood boring. The alternate round is used with electric drills. The double twist is not as strong as the solid center, but removes chips rapidly. Finally, single twist provide more space for wood chips and are suited for boring deep holes

16. Drill sizes are commonly designated in

A) fractions of an inch
B) fractions of a centimeter
C) whole inches
D) whole inches

17. The ends of the drill bit are called

A) the shank and the grip
B) the bit and the edge
C) the grip and the cutting edge
D) the shank and the cutting edge

18. The best drill option if you need to bore deep holes is the

A) single twist
B) double twist
C) solid center
D) alternate round

Answers to Practice Test

1) A	**2)** B	**3)** C	**4)** D	**5)** D	**6)** B
7) C	**8)** D	**9)** D	**10)** A	**11)** A	**12)** B
13) B	**14)** D	**15)** C	**16)** A	**17)** D	**18)** A



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